Generating Piano Music with Transformers: A Comparative Study of Scale, Data and Metrics

NEURAL INFORMATION PROCESSING SYSTEMS

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Methodology

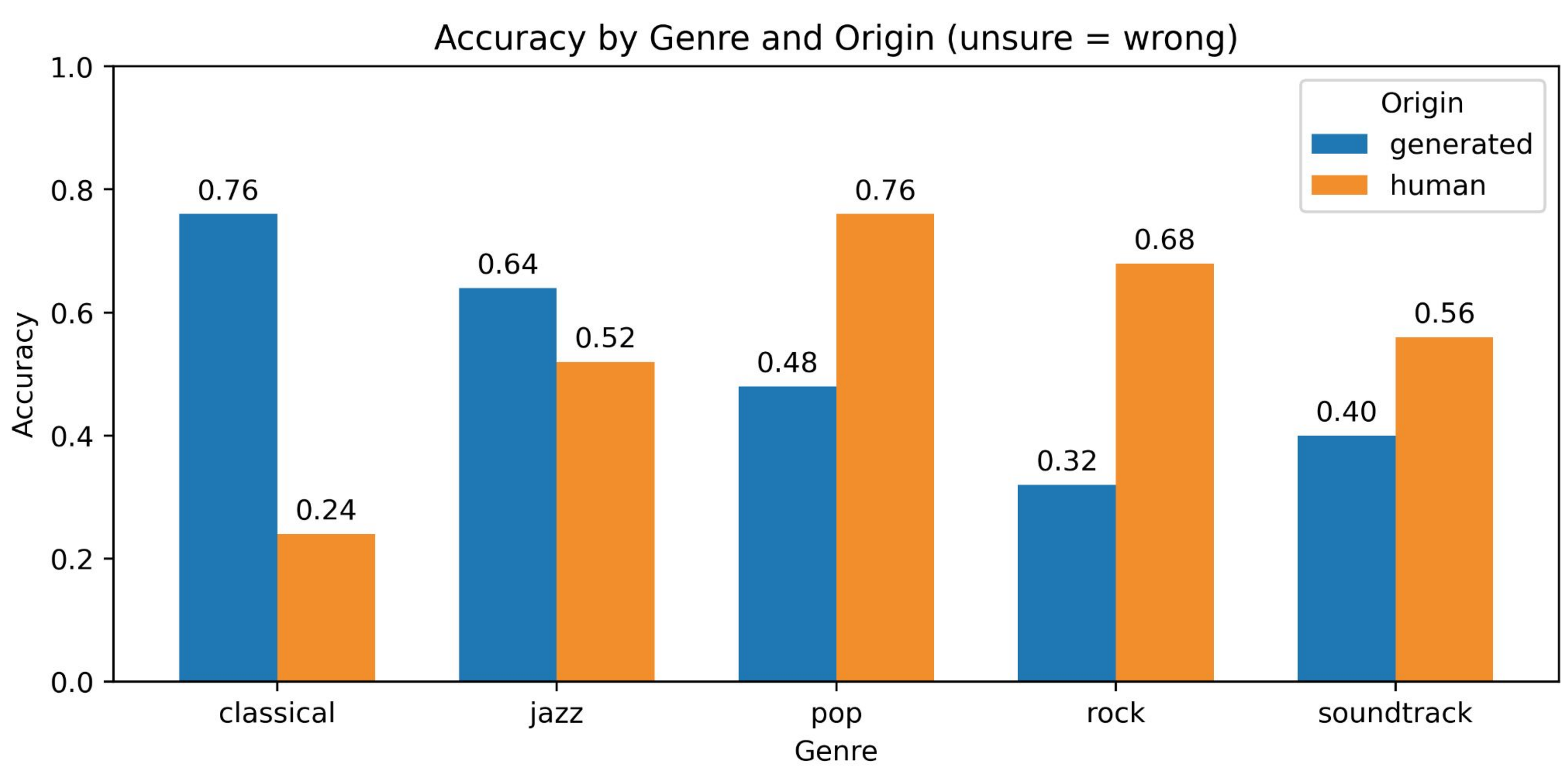
We examined how model scale, datasets, training strategies and metrics affect symbolic piano music generation. **MIDI** data from **MAESTRO** and **Aria** are split into chunks of 1024 tokens and augmented through variations in pitch, velocity, duration, and tempo. Our **REMI** vocabulary contains 485 base tokens, which we extend using **byte-pair encoding** to learn a 30K-token vocabulary. We use **Fréchet Music Distance (FMD)**, **Kullback–Leibler divergence (KLD)**, and **Overlapping Area (OA)** to compare feature-level distributions of generated and training samples.

Model Scaling & Moonbeam Benchmark

Model	Mean S.	Pleas.	Auth.	Nov.	FMD↓	KLD↓	OA↑	PPL↓
62M	2.84	2.85	2.78	2.89	210.01	0.36	62.85%	53.65
155M	2.86	2.89	2.66	3.03	187.53	0.38	68.46%	21.32
439M	**3.22**	**3.30**	**3.02**	**3.35**	**176.40**	**0.25**	71.60%	4.82
950M	3.17	3.27	2.98	3.27	176.80	0.29	68.33%	**2.78**
MB 839M	2.91	2.90	2.74	3.08	194.70	0.51	**82.21%**	–

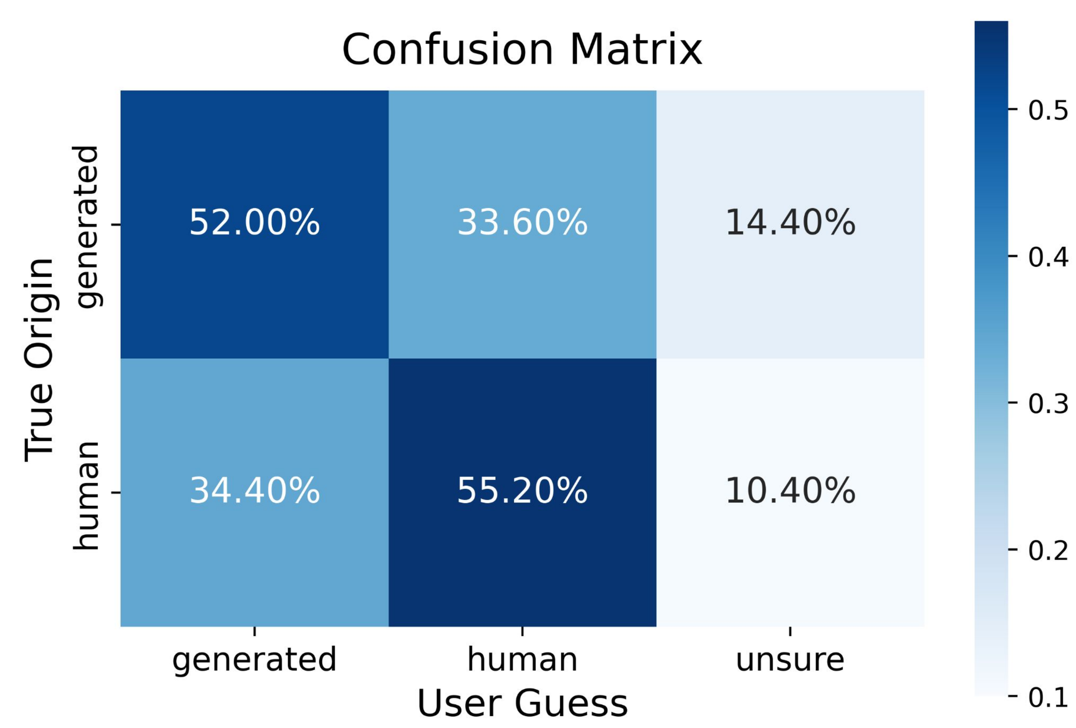
We trained four **Mistral**-based models on MAESTRO: Musical quality generally improves with model size despite increased **overfitting**. Later checkpoints sometimes yield better even as validation loss worsens, indicating that mild overfitting may help models capture high-level musical structure. However, the 950M model underperforms the 439M model on KLD, OA, and subjective ratings at the final checkpoint. For comparison, we fine-tuned the **Moonbeam** foundation model on MAESTRO.

Turing-like Listening Study

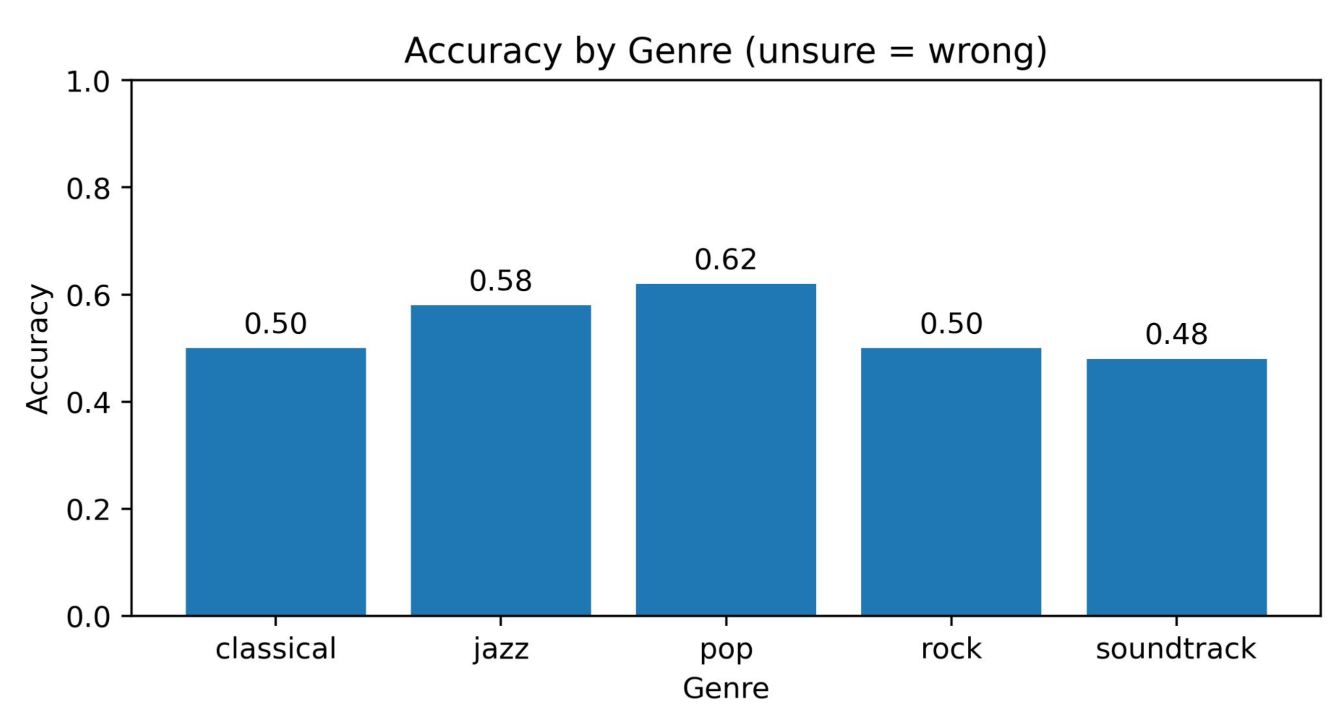


The genre-conditioned 950M model was evaluated through a musical **Turing-like** test across five genres from Aria. For each genre, five generated and five human-composed sequences were selected from a pool of thirty samples. Excluding "unsure" responses, participants achieved 61.2% accuracy. **Precision**, **recall**, and **F1**-scores were similar for human and generated samples, and the **confusion matrix** shows roughly a third of samples misclassified, indicating many generated sequences were perceptually similar to human music.

Class	**Precision**	**Recall**	**F1-score**
Human	62.16%	55.20%	58.47%
Generated	60.19%	52.00%	55.79%



Accuracy by genre ranged from **48%** (soundtrack) to **62%** (pop), with biases: classical and jazz leaned "generated," while pop, rock, and soundtrack leaned "human". Overall, responses were **44.4% human**, **43.2% generated**, and **12.4% unsure**, showing slight bias. Unsure rates were slightly higher for generated sequences (**14.4%** vs. **10.4%**), suggesting confusion remained high across genres.



Transfer Learning: Fine-Tuning Aria-Pretrained Models on MAESTRO

Aria is large but noisy and MAESTRO is small but high quality: We apply transfer learning to combine both their strengths. Models pretrained on Aria are fine-tuned on MAESTRO either by freezing the first eight blocks or full fine-tuning. Both variants outperform the MAESTRO-only baseline on subjective and most objective metrics, suggesting that allowing all layers to adapt is beneficial, likely due to the genre gap between the multi-genre pre-training data and classical MAESTRO.

Model	Mean S.	Pleas.	Auth.	Nov.	FMD↓	KLD↓	OA↑
155M	2.85	2.80	2.80	2.95	187.53	0.38	68.46%
155M-F-P	3.09	3.13	2.95	3.20	193.45	0.34	70.93%
155M-F-F	**3.25**	**3.30**	**3.10**	**3.35**	**187.34**	**0.30**	**72.39%**

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Check out music samples in our repository: